

Answer All Three Questions

The numbers in square brackets show the provisional allocation of maximum marks per question or part of question.

A module materials booklet, including physical constants if applicable, is attached at the end of the paper.

[Part marks]

1. (a) The unperturbed Hamiltonian $\hat{H}_0$ of a pair of quantum harmonic oscillators (QHOs) is given by:

$$\hat{H}_0 = \frac{\hat{p}_z^2}{2m} + \frac{1}{2}m\omega_z^2\hat{z}^2 + \frac{\hat{p}_y^2}{2m} + \frac{1}{2}m\omega_y^2\hat{y}^2.$$

Write the Hamiltonian above in terms of raising/lowering operators $\hat{a}_{\pm}^{(z)}$ and $\hat{a}_{\pm}^{(y)}$, defining all quantities carefully.

Give an expression for the eigenvalues corresponding to the eigenstates $|n_y, n_z\rangle$ of $\hat{H}_0$.

[5]

- (b) The above oscillators are now subject to an additional potential such that the new Hamiltonian takes the form:
 $\hat{H} = \hat{H}_0 + ig_y(\hat{a}_+^{(y)} - \hat{a}_-^{(y)}) + ig_z(\hat{a}_+^{(z)} - \hat{a}_-^{(z)}).$

Represent the Hamiltonian $\hat{H}$ in matrix form using the eigenbasis of $\hat{H}_0$, restricting yourself to quantum numbers $n_y, n_z \leq 1$, and for the case $\omega_y = \omega_z = \omega$.

[7]

- (c) For an arbitrary QHO, show that $[\hat{a}_+\hat{a}_-, \hat{a}_+] = \hat{a}_+$ and $[\hat{a}_+\hat{a}_-, \hat{a}_-] = -\hat{a}_-.$

[4]

Hence or otherwise, for the pair of QHOs above, obtain an uncertainty relation between the energy and $\hat{P}_z$; in other words, given $\Delta\hat{H}\Delta\hat{P}_z \geq C_z$, obtain C_z .

[4]

CONTINUED

[Part marks]

2. (a) Angular momentum operators are associated with raising and lowering operators defined as $\hat{J}_{\pm} = \hat{J}_x \pm i\hat{J}_y$.

Let $|j m\rangle$ denote the simultaneous eigenvector of both $\hat{J}^2$ and $\hat{J}_z$, the corresponding eigenvalue equations being:

$$\hat{J}^2|j m\rangle = \alpha|j m\rangle \text{ and } \hat{J}_z|j m\rangle = \beta|j m\rangle.$$

Show that $[\hat{J}_z, \hat{J}_{\pm}] = \hbar\hat{J}_{\pm}$.

[3]

- (b) Use this commutator to show that the operation of $\hat{J}_{+}$ on the eigenvalue equation for $\hat{J}_z$ generates a series of eigenstates with eigenvalues equal to $\beta + n\hbar$ where n is an integer.

[6]

- (c) Two particles have angular momentum quantum numbers $j_1 = 1$ and $j_2 = 1/2$ respectively. They experience a Hamiltonian:

$$\hat{H} = C_{12}\mathbf{J}_1 \cdot \mathbf{J}_2 + A(\hat{J}_{1z} + \hat{J}_{2z})^2$$

where C_{12} and A are constants. Give the interaction term $\mathbf{J}_1 \cdot \mathbf{J}_2$ in terms of raising and lowering operators $\hat{J}_{\pm 1}$ and $\hat{J}_{\pm 2}$ and any other operators you may require.

Hence represent the above Hamiltonian as a matrix, using the basis $|j_1 m_1\rangle|j_2 m_2\rangle \equiv |m_1 m_2\rangle$ being careful to define all terms.

[7]

- (d) As an alternative basis, one may construct total angular momentum states $|JM; j_1 j_2\rangle$. Give all the values of J and M you would expect for these states. Show that the $|JM; j_1 j_2\rangle$ are eigenstates of $\hat{H}$ and give the eigenvalues.

[4]

CONTINUED

[Part marks]

3. (a) A spin-half particle in a magnetic field oriented along the x-direction has a Hamiltonian

$$\hat{H} = \hat{H}_0 = A\hat{\sigma}_x,$$

where $A > 0$. An additional magnetic field is applied in the z-direction, and the Hamiltonian changes to

$$\hat{H} = \hat{H}_0 + \hat{W} = A\hat{\sigma}_x + \alpha\hat{\sigma}_z,$$

where $\alpha > 0$. Initially $\alpha \ll A$.

- i. Given that the eigenvectors of the σ_x matrix are

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

compute the eigenvalues of $\hat{H}_0$. [2]

- ii. Use perturbation theory to calculate both the first order and the second order energy shifts. [5]

- iii. If the strength of the additional field is significantly increased so that $\alpha \gg A$, can we still use perturbation theory to estimate the eigenvalues in this limit? Explain your answer. [2]

- (b) A three-level system has a Hamiltonian $\hat{H} = \hat{H}_0 + \hat{W}$ where $\hat{H}_0$ and $\hat{W}$ are represented by the following matrices:

$$\hat{H}_0 = \hbar \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad \hat{W} = \beta \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 2i \\ 3 & -2i & 0 \end{bmatrix}$$

- i. Identify the eigenenergies of $\hat{H}_0$ and the corresponding orthonormal eigenvectors. Which energy levels are degenerate? [3]

- ii. Assuming $\beta > 0$ and $\beta \ll \hbar$, compute the first order energy shifts to the ground state and excited states. [8]

END OF EXAMINATION PAPER

PHAS0042 Materials Booklet 2022-23

1 Formal Quantum Mechanics

We often represent state vectors in an orthonormal basis, $|\phi_n\rangle$, $n = 1, \dots, d$ where $\langle\phi_n|\phi_m\rangle = \delta_{n,m}$.

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle$$

We can represent $|\psi\rangle$ with a column vector

$$|\psi\rangle \rightarrow \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{bmatrix}$$

Similarly, any bra vector can be represented as a row vector

$$\langle\psi| = \sum_n c_n^* \langle\phi_n| \quad \langle\psi| \rightarrow [c_1^* \quad c_2^* \quad c_3^* \quad \dots]$$

and any operator $\hat{A}$ can be represented by a matrix A , where the matrix elements of A are given by

$$A_{j,k} = \langle\phi_j|\hat{A}|\phi_k\rangle.$$

2 The Quantum Harmonic Oscillator

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi_n(x) = E_n \psi_n(x).$$

Its (orthonormal) eigenfunctions are:

$$\psi(x) = N_n(\alpha) H_n(\alpha x) e^{-\alpha^2 x^2/2}.$$

where $H_n(\alpha x)$ is Hermite polynomial of degree n , parity $(-1)^n$ while

$$N_n = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{1/2}, \text{ and } E_n = \left(n + \frac{1}{2} \right) \hbar\omega$$

$$\text{and } [\hat{x}, \hat{p}] = i\hbar$$

Raising and lowering operators

$$\hat{a}_+ = \frac{1}{\sqrt{2}} \left[\alpha \hat{x} - \frac{i}{\hbar\alpha} \hat{p} \right] \quad \text{and} \quad \hat{a}_- = \frac{1}{\sqrt{2}} \left[\alpha \hat{x} + \frac{i}{\hbar\alpha} \hat{p} \right], \text{ where } \alpha = (m\omega/\hbar)^{1/2}$$

$$\text{For arbitrary operators, } [\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}] \hat{C} + \hat{B} [\hat{A}, \hat{C}]$$

Key equations

$$\hat{a}_+^\dagger = \hat{a}_-; \quad \hat{a}_-^\dagger = \hat{a}_+; \quad [\hat{a}_-, \hat{a}_+] = 1.$$

$$\hat{H} = \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) \hbar \omega; \quad \hat{N}|n\rangle = n|n\rangle$$

$$\hat{a}_-|n\rangle = \sqrt{n}|n-1\rangle; \quad \langle k|\hat{a}_-|n\rangle = \sqrt{n}\delta_{k,n-1} \quad (1)$$

$$\hat{a}_+|n\rangle = \sqrt{n+1}|n+1\rangle; \quad \langle k|\hat{a}_+|n\rangle = \sqrt{n+1}\delta_{k,n+1}. \quad (2)$$

$$(\Delta A)(\Delta B) \geq \frac{1}{2} | \langle [\hat{A}, \hat{B}] \rangle |$$

where:

$$(\Delta A)^2 = \langle \psi | (\hat{A} - \langle \hat{A} \rangle)^2 | \psi \rangle$$

3 & 4 Angular momentum and spin

The time-independent Schrödinger equation is

$$\hat{L}^2 Y_{lm}(\theta, \phi) = l(l+1)\hbar^2 Y_{lm}(\theta, \phi) \text{ and } \hat{L}_z Y_{lm}(\theta, \phi) = m\hbar Y_{lm}(\theta, \phi)$$

$$[\hat{L}_i, \hat{L}^2] = 0 \text{ for } i = x, y, z \quad [\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z \text{ and cyclic permutations}$$

We convert cartesian coordinates to spherical polars:

$$x = r \sin \theta \cos \phi; \quad y = r \sin \theta \sin \phi \quad z = r \cos \theta$$

Angular momentum with raising and lowering operators

$$[\hat{J}_i, \hat{J}_j] = i\hbar J_k \text{ for cyclic permutations of } i, j, k \equiv x, y, z$$

$$\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y$$

Key equations

$$(\hat{J}_\pm)^\dagger = \hat{J}_\mp; \quad \hat{J}_\pm \hat{J}_\mp = \hat{J}^2 - \hat{J}_z^2 \pm \hbar \hat{J}_z$$

$$[\hat{J}_\pm, \hat{J}_\mp] = \pm 2\hbar \hat{J}_z; \quad [\hat{J}_z, \hat{J}_\pm] = \pm \hbar \hat{J}_\pm$$

$$\hat{J}^2 |jm\rangle = j(j+1)\hbar^2 |jm\rangle \quad \text{and } \hat{J}_z |jm\rangle = m\hbar |jm\rangle$$

$$\hat{J}_- |jm\rangle = \sqrt{j(j+1) - m(m-1)} \hbar |jm-1\rangle \quad (3)$$

$$\hat{J}_+ |jm\rangle = \sqrt{j(j+1) - m(m+1)} \hbar |jm+1\rangle \quad (4)$$

The Pauli matrices take the form:

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The eigenstates of the operator $\hat{n} \cdot \hat{\sigma}$ may be written:

$$|+\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} \sin \frac{\theta}{2} \\ -\cos \frac{\theta}{2} e^{i\phi} \end{pmatrix}$$

Adding Angular momenta

For a total angular momentum $\hat{\mathbf{J}} = \hat{\mathbf{J}}_1 + \hat{\mathbf{J}}_2$, we have eigenstates

$$\hat{\mathbf{J}}^2 |JM\rangle = J(J+1)\hbar^2 |JM\rangle \quad \text{and} \quad \hat{J}_z |JM\rangle = M\hbar |JM\rangle$$

that are constructed using Clebsch Gordan coefficients $\langle j_1 m_1 j_2 M - m_1 | JM; j_1 j_2 \rangle$:

$$|JM\rangle \equiv |JM; j_1 j_2\rangle = \sum_{m_1} \langle j_1 m_1 j_2 M - m_1 | JM; j_1 j_2 \rangle |j_1 m_1\rangle |j_2 M - m_1\rangle,$$

We can show that

$$\hat{\mathbf{J}}^2 = \hat{\mathbf{J}}_1^2 + \hat{\mathbf{J}}_2^2 + 2\hat{\mathbf{J}}_1 \cdot \hat{\mathbf{J}}_2 = \hat{\mathbf{J}}_1^2 + \hat{\mathbf{J}}_2^2 + 2J_{z1}J_{z2} + \hat{J}_{1+}\hat{J}_{2-} + \hat{J}_{1-}\hat{J}_{2+}$$

A table of **Clebsch-Gordan coefficients** is on the last page of this document.

Source: https://en.wikipedia.org/wiki/Table_of_Clebsch-Gordan_coefficients

5 Quantum Mechanics with Many Particles

CHSH-Bell inequality

$$E(C) = E(a_0 b_0) + E(a_0 b_1) + E(a_1 b_0) - E(a_1 b_1) \leq 2$$

Slater Determinant

$$\Psi_{n_1 n_2 \dots n_N}^{(A)}(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_{n_1}(r_1) & \phi_{n_1}(r_2) & \cdots & \phi_{n_1}(r_N) \\ \phi_{n_2}(r_1) & \phi_{n_2}(r_2) & \cdots & \phi_{n_2}(r_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{n_N}(r_1) & \phi_{n_N}(r_2) & \cdots & \phi_{n_N}(r_N) \end{vmatrix}$$

6 Approximate Methods

Perturbation Theory - nondegenerate form

We have a Hamiltonian of the form:

$$\hat{H} = \hat{H}_0 + \lambda \hat{W}$$

where $\hat{H}_0$ is non-degenerate and $\hat{H}_0 |\psi_n^{(0)}\rangle = E_n^{(0)} |\psi_n^{(0)}\rangle$.

We form power series expansions:

$$E_n = \sum_{j=0}^{\infty} \lambda^j E_n^{(j)} = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \quad |\psi_n\rangle = \sum_{j=0}^{\infty} \lambda^j |\psi_n^{(j)}\rangle = |\psi_n^{(0)}\rangle + \lambda |\psi_n^{(1)}\rangle + \lambda^2 |\psi_n^{(2)}\rangle + \dots$$

The first order correction to the eigenvalues are:

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{W} | \psi_n^{(0)} \rangle.$$

The first order correction to the eigenvectors is:

$$|\psi_n^{(1)}\rangle = \sum_{k \neq n} \frac{W_{kn}}{E_n^{(0)} - E_k^{(0)}} |\psi_k^{(0)}\rangle.$$

where here $W_{\ell n} = \langle \psi_\ell^{(0)} | \hat{W} | \psi_n^{(0)} \rangle$.

The second order correction to the eigenvalues is:

$$E_n^{(2)} = \sum_{k \neq n} \frac{|W_{kn}|^2}{(E_n^{(0)} - E_k^{(0)})} = \sum_{k \neq n} \frac{|\langle \psi_k^{(0)} | \hat{W} | \psi_n^{(0)} \rangle|^2}{(E_n^{(0)} - E_k^{(0)})}.$$

Perturbation Theory - degenerate form

We have a Hamiltonian of the form:

$$\hat{H} = \hat{H}_0 + \lambda \hat{W}$$

where $\hat{H}_0$ is degenerate and we label its degenerate eigenstates $\hat{H}_0 |\chi_{n\mu}\rangle = E_n |\chi_{n\mu}\rangle$.

Again, we expand eigenvalues and eigenvectors as a power series

$$|\psi_{n\mu}\rangle = |\psi_{n\mu}^{(0)}\rangle + \lambda |\psi_{n\mu}^{(1)}\rangle + \lambda^2 |\psi_{n\mu}^{(2)}\rangle + \dots, \quad E_{n\mu} = E_n^{(0)} + \lambda E_{n\mu}^{(1)} + \lambda^2 E_{n\mu}^{(2)} + \dots$$

We compute the first order eigenvalue correction and identify the zeroth order eigenvector term by solving a matrix equation. For example, for a 2-fold degenerate eigenvalue of $\hat{H}_0$, it has the form:

$$\begin{pmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{pmatrix} \begin{pmatrix} c_{j1} \\ c_{j2} \end{pmatrix} = E_{nj}^{(1)} \begin{pmatrix} c_{j1} \\ c_{j2} \end{pmatrix}.$$

for $j = 1, 2$ where here $W_{ij} = \langle \chi_{ni} | \hat{W} | \chi_{nj} \rangle$ and where

$$|\psi_{nj}^{(0)}\rangle = c_{j1} |\chi_{n1}\rangle + c_{j2} |\chi_{n2}\rangle$$

Variational Method

The variational principle states that

$$\frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0.$$

where $|\psi\rangle$ is any state, which does not have to be normalised.

The variational method allows us to derive an upper bound of (and hence an estimate of) the ground state energy by minimising the energy of a parameterised trial solution $|\psi_{\vec{\alpha}}\rangle = |\psi_T(\alpha_1, \alpha_2, \dots, \alpha_s)\rangle$.

We thus compute

$$E(\alpha_1, \alpha_2, \dots, \alpha_s) = E(\vec{\alpha}) = \langle \hat{H} \rangle_{\vec{\alpha}} = \frac{\langle \psi_{\vec{\alpha}} | \hat{H} | \psi_{\vec{\alpha}} \rangle}{\langle \psi_{\vec{\alpha}} | \psi_{\vec{\alpha}} \rangle}.$$

and use calculus to determine the minimum value of $E(\vec{\alpha})$.

CLEBSCH-GORDAN COEFFICIENTS

$$j_1 = \frac{1}{2}, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = 1$$

m_1, m_2	j	1
$\frac{1}{2}, \frac{1}{2}$	1	1

$$m = -1$$

m_1, m_2	j	1
$-\frac{1}{2}, -\frac{1}{2}$	1	1

$$m = 0$$

m_1, m_2	j	1	0
$\frac{1}{2}, -\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$-\frac{1}{2}, \frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$j_1 = 1, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = \frac{3}{2}$$

m_1, m_2	j	$\frac{3}{2}$	$\frac{1}{2}$
$1, \frac{1}{2}$	1	1	

$$m = \frac{1}{2}$$

m_1, m_2	j	$\frac{3}{2}$	$\frac{1}{2}$
$1, -\frac{1}{2}$	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{2}{3}}$
$0, \frac{1}{2}$	$\sqrt{\frac{2}{3}}$	$\sqrt{\frac{2}{3}}$	$-\sqrt{\frac{1}{3}}$

For brevity, solutions with $M < 0$ and $j_1 < j_2$ are omitted. They may be calculated using the simple relations

$$\langle j_1, j_2; m_1, m_2 \mid j_1, j_2; J, M \rangle = (-1)^{J-j_1-j_2} \langle j_2, j_1; -m_1, -m_2 \mid j_1, j_2; J, -M \rangle.$$

and

$$\langle j_1, j_2; m_1, m_2 \mid j_1, j_2; J, M \rangle = (-1)^{J-j_1-j_2} \langle j_2, j_1; m_2, m_1 \mid j_2, j_1; J, M \rangle.$$

$$j_1 = \frac{3}{2}, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = 2$$

m_1, m_2	j	2
$\frac{3}{2}, \frac{1}{2}$	1	1

$$m = 1$$

m_1, m_2	j	2	1
$\frac{3}{2}, -\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{3}{4}}$
$1, \frac{1}{2}$	$\sqrt{\frac{3}{4}}$	$\sqrt{\frac{3}{4}}$	$-\frac{1}{2}$

$$m = 0$$

m_1, m_2	j	2	1
$\frac{1}{2}, -\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$1, \frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$j_1 = 1, j_2 = 1 \quad [\text{edit}]$$

$$m = 2$$

m_1, m_2	j	2
1, 1	1	1

$$m = 1$$

m_1, m_2	j	2	1
1, 0	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
0, 1	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$m = 0$$

m_1, m_2	j	2	1	0
1, -1	$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$
0, 0	$\sqrt{\frac{2}{3}}$	$\sqrt{\frac{2}{3}}$	0	$-\sqrt{\frac{1}{3}}$
-1, 1	$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{6}}$	$-\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$